Studio North: customer economics lab

Are we moving in the right direction as a company?

Compare acquisition efficiency, contribution value and cash recovery across cohorts.

A monthly creative-work membership for independent professionals: templates save time, guided learning builds skills, and workshops add value.

24 joining cohorts: January 2024–December 2025.
Monthly renewals; cutoff January 1, 2026 (exclusive).

CRM	Renewals
Transactions	Product events
Feedback	Campaign spend
Source credit	Assignments

How to read the slide colors

Concept	Definitions and distinctions to use when interpreting the data.
Case application	Studio North and its connected customer records.
Evidence interpretation	Observed results, predictions, and what each can establish.
Group decision	One focused judgment, discussed with your group.

Three ideas guide the entire lab

1. Connected customer evidence

The Universal Customer View classifies each dataset's fields as Identifiers, Engagement, Relationship or Perception. It reveals what is recorded and what is missing.

2. Customer economics over time

Joining cohorts distinguish customer age from calendar time. Retention, contribution and acquisition cost determine whether relationships create value.

3. Incremental marketing value

A randomized campaign comparison asks what changed because of the campaign. A high customer LTV alone cannot answer that question.

AI proposes the work. Saved SQL, data checks, held-out outcomes and your judgment make the work reviewable.

Every dataset gets its own Universal Customer View

Example: people.csv contains one record per person, including prospects who never buy.

Table 1. Field classification for people.csv

Identifiers	Engagement	Relationship	Perception
person_id age_band household_type region	—	record_created_at attributes_recorded_at	primary_need

The stated need is a reported intention. Profile timestamps document when the relationship information became available.

Repeat this field-level table for all eight exports, then open the Archify map of keys, grains and connections.

Group decision 1: who needs which onboarding route?

Studio North can offer a ready-to-use template or a guided lesson. Which recorded grouping should inform these onboarding routes?

Table 2. Two candidate grouping rules

Grouping	Two recorded groups	What the field describes
Age band	18–34 / 35+	A demographic characteristic.
Stated need	Save time / Build skill	The goal stated before acquisition.

Choose one grouping and explain why it is relevant to this onboarding objective.

Use the opening business map to identify the first-value decision. Then compare retention and contribution within groups.

Turn the connected view into a reproducible query

Inspect raw views

Keep source columns and row meanings.

Define the customer clock

First paid month sets cohort age index 0.

Build SQL outputs

Save the exact SQL and ordered results.

Reconcile the totals

No duplicate money; no missing controls.

Figure 1. From source views to reconciled cohort evidence

Prompt to try

“Show the SQL that calculates retained customers and contribution by joining month and customer age. Explain the denominator and preserve unobserved future ages as missing.”

A SQL script makes a calculation inspectable; the data definitions determine whether it answers the right question.

A cohort follows the same original customers

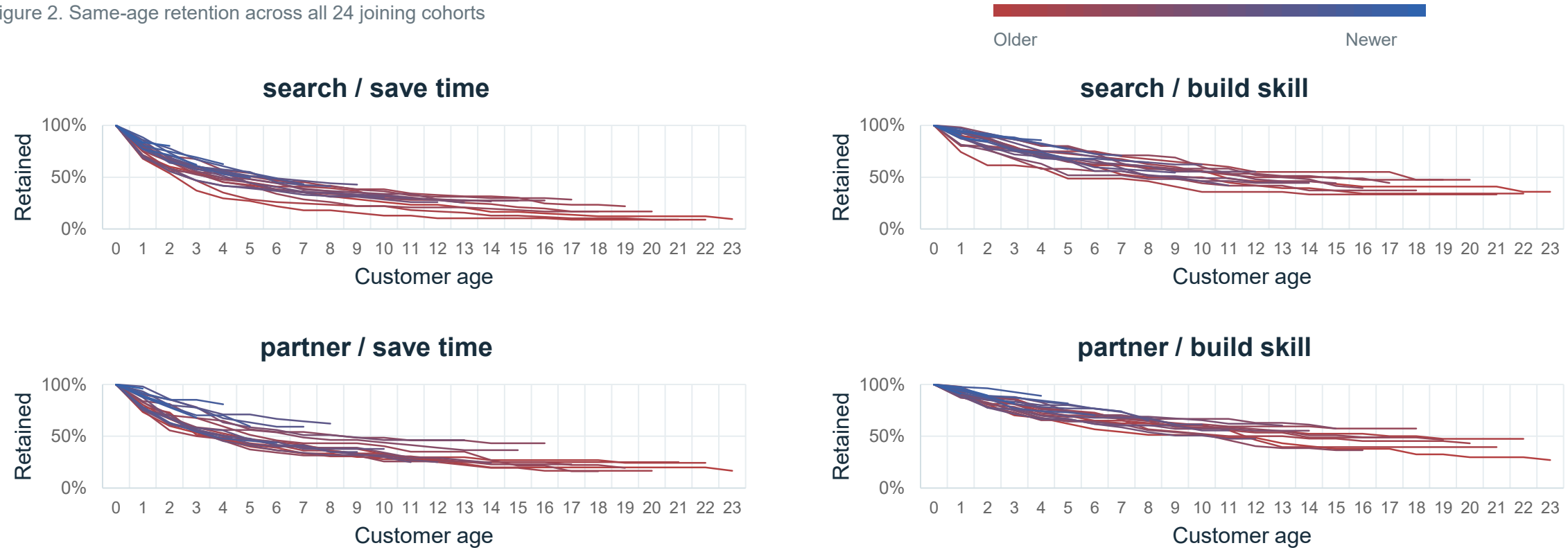
Table 3. Retained share by joining cohort and customer age (months)

Cohort	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
2024-01	100%	80%	70%	63%	55%	51%	47%	45%	42%	40%	39%	35%	34%	31%	28%	27%	26%	26%	24%	24%	23%	23%	22%	20%
2024-02	100%	78%	70%	56%	49%	46%	42%	38%	37%	36%	33%	32%	30%	29%	28%	27%	27%	26%	26%	25%	25%	25%	25%	—
2024-03	100%	78%	66%	59%	51%	44%	42%	40%	38%	36%	34%	32%	29%	27%	25%	25%	25%	24%	24%	24%	24%	24%	—	—
2024-04	100%	84%	72%	67%	62%	59%	55%	52%	49%	46%	42%	40%	38%	37%	35%	34%	32%	31%	30%	30%	29%	—	—	—
2024-05	100%	78%	67%	65%	62%	60%	57%	54%	50%	47%	45%	42%	40%	39%	38%	38%	35%	34%	34%	32%	—	—	—	—
2024-06	100%	82%	73%	65%	57%	53%	49%	46%	44%	41%	39%	37%	37%	37%	34%	32%	31%	30%	30%	—	—	—	—	—
2024-07	100%	85%	77%	72%	63%	60%	56%	53%	52%	49%	45%	42%	41%	40%	39%	38%	38%	36%	—	—	—	—	—	—
2024-08	100%	80%	72%	65%	60%	56%	52%	51%	48%	46%	45%	42%	40%	38%	37%	35%	35%	—	—	—	—	—	—	—
2024-09	100%	87%	78%	72%	66%	63%	59%	54%	50%	50%	49%	48%	43%	42%	42%	40%	—	—	—	—	—	—	—	—
2024-10	100%	82%	71%	63%	57%	53%	50%	47%	46%	44%	42%	40%	39%	38%	37%	—	—	—	—	—	—	—	—	—
2024-11	100%	87%	80%	75%	67%	60%	56%	52%	50%	48%	46%	45%	44%	42%	—	—	—	—	—	—	—	—	—	—
2024-12	100%	81%	73%	63%	57%	54%	50%	48%	44%	42%	41%	39%	38%	—	—	—	—	—	—	—	—	—	—	—
2025-01	100%	87%	73%	64%	59%	54%	49%	47%	45%	44%	41%	38%	—	—	—	—	—	—	—	—	—	—	—	—
2025-02	100%	82%	70%	64%	60%	56%	54%	50%	49%	47%	47%	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-03	100%	85%	74%	69%	64%	61%	57%	55%	51%	49%	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-04	100%	90%	78%	73%	69%	67%	63%	60%	57%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-05	100%	88%	78%	72%	68%	64%	60%	57%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-06	100%	87%	79%	70%	65%	61%	58%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-07	100%	92%	82%	76%	71%	66%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-08	100%	90%	87%	83%	78%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-09	100%	89%	82%	76%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-10	100%	89%	84%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-11	100%	89%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-12	100%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Read across for one cohort over time, or down for different cohorts at the same age. Future ages stay missing.

Group decision 2: is improvement visible within groups?

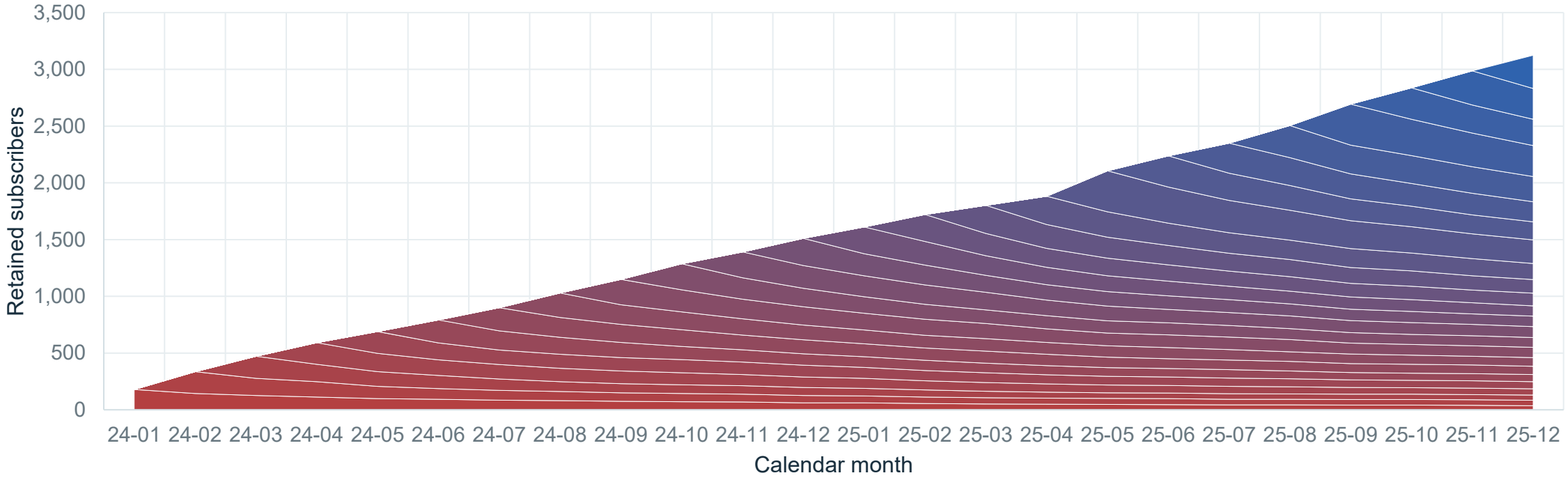
Figure 2. Same-age retention across all 24 joining cohorts



Do newer cohorts retain better at the same age? Is the progress broad or concentrated?

Joining cohorts accumulate into the company

Figure 3. Retained subscribers by joining cohort and calendar month



Each band is one joining cohort. The height is the retained customer base, not cumulative acquisitions.

Group decision 3: is one month enough to decide?

Each campaign randomized 250 prospects to treatment and 250 to its own control. Follow-up lasts about one month.

Table 4. Experimental contribution after acquisition spending

Test	Follow-up	Incremental contribution after media / prospect	Approx. 95% interval
search / save time	31 days	$1.91	$0.03 to $3.78
partner / build skill	30 days	-$0.34	-$2.33 to $1.65

Would you scale, stop, or keep measuring? What evidence would change your decision?

Cohort payback takes months. Predicted customer LTV cannot replace longer experimental follow-up.

Model renewal and revenue before forecasting LTV

Retention model

A hierarchical shifted Beta-Geometric model represents persistent differences in churn risk. Cohort and group information share strength where histories are short.

Revenue model

A Gamma model describes positive revenue among retained customers. Combine this forecast with retention, variable costs and observed contribution.

Validation

Compare chronological holdout predictions with a simple baseline and an Orduz-style neural network. Keep uncertainty visible; a fitted curve is not a guarantee.

Value the first 12 monthly receipts for every cohort. Net LTV subtracts CAC; cash payback tracks undiscounted recovery.

Run the locked NumPyro/JAX models on CPU with uv. Describe the model before showing its results.

Hand your team a decision and a reproducible report

Are successive cohorts becoming more valuable relative to acquisition cost?

Table 5. The executive handoff

Report component	What your team should receive
Executive summary	The business, a supported verdict, a proposed action and what would change it.
Observed and forecast evidence	Same-age retention, revenue, contribution, CAC and cash payback.
Model appendix	Model story, parameters, priors, uncertainty and chronological validation.
Data and reproducibility appendix	UCV tables, Archify key map and saved SQL; saved manifests accompany the report.

Bonus challenge: forecast 2026 outcomes for these 24 cohorts with another CPU-capable model. Submit for instructor-held evaluation.

Open with the business and decision; explain extrapolation early. Keep model details and reproducibility in the appendix.